# Unit V

## Sustainable Design and Product Life Cycle Management

**Design for robustness, safety, and reliability, Design for environment and sustainability, Basic life cycle assessment methods, Weighted sum assessment method, Design failure mode effect analysis, Product life cycle management concepts, Product data management and workflow, Local, regional and global design considerations.**

### Design for Robustness

- Robustness is a property of systems that enable the survive unforeseen or unusual circumstances.
- A robust design is one whose performance is insensitive to variations in the manufacturing processes by which it has been made or in the environment in which it operates. It is basic tenet of quality that variations of all kinds are enemy of quality and guiding principle of quality is to reduce variation. The methods used to achieve robustness are termed as robust design. These are basically the work of a Japanese engineer Genichi Taguchi and his co-workers, adopted by manufacturing companies worldwide. They employed a set of statistically designed experiments by which alternative designs are generated and analysed for their sensitivity to variation. The prototyping design step is the place where design for robustness methods are applied to critical to quality generators.

### Tools for Robust Design

- Taguchi Methods design of experiments and multiple regression analysis are some of the important tools used for robust design to produce high quality products quickly and at low cost.

### Design for Safety

- Organization that eliminate or reduce hazards by making design or engineering changes generally improve their workplace. Safety and health and save money in the long run.
- **Design for safety** is a planned, disciplined and systematic approach for preventing or reducing accidents through out the life cycle of a system.
- Primary concern in the management of risks.
- Risk identification, evaluation, elimination and control.
- Through analysis, design and management.

### Methods

- There are four main methods of design for safety:

a) **Minimize**: Reducing the amount of hazardous material present at any one time.

b) **Substitute**: Replacing one material with less hazardous one.

e.g.: Cleaning with water and detergent rather than a flammable solvent.

c) **Moderate**: Reducing the strength of an effect.

e.g.: Using material in dilute form rather than concentrated form.

d) **Simplify**: Designing out problems rather than adding additional equipment or features to dealt with them.

**Design for Reliability (DFR)**

- Reliability professionals these days are more and more familiar with the term Design For Reliability (DFR).
- Design for reliability is an engineering discipline that refers to the process of designing reliability into the products. This process begin from the very early stages of design and should be integrated into every stage of this process.
- This process will enable setting up goals at the beginning of design process, driven by the design team with the reliability team acting as a mentors.

**DFR Phases or Key Activities**

- The major phase of a DFR program typically contain the following steps.

1.Identify 2.Design 3.Analyze 4.Verify 5.Validate 6.Monitor and control

**Identify Phase**

- The goal of identify phase is to quantitatively define the reliability requirements for product as well as the end user environmental/ usage conditions.
- Review customer expectations and how to translate them into engineering metrics (e.g. - serve 15 years life).
- Identify technology limitation (e.g. - battery, optics, specific components etc.) and relevant validation strategy.
- In order to identify reliability, requirement, bench marking, quality function deployment, ...etc. tools being used.

**Design Phase**

- The design phase is the stage, where specific design activities begins such as circuit layout, mechanical drawing, component/supplier selection etc. In this stage a clearer picture about what the product supposed to do starts developing.

- Also following can be achieved
  - More specific reliability requirements are defined.
  - The more design/application change, the more reliability risks are introduced.
  - Tolerance evaluation.
  - Better understanding of customer specifications.
  - Failure Modes and Effect Analysis (FMEA)

**Analyze Phase**

- In the analyze phase, eliminates the product reliability, often with a rough first cut estimate, early in design phase. It is important that at this phase to address all the potential sources of product failure. This requires close cooperation between reliability engineer and the design team.
- Some tools used in the Analyze phase are.

1.Finite Element Analysis (FEA) 2.Physics of failure

3.Reliability prediction 4.Warranty Analysis

5.Design review by failure mode…etc.

**Verify Phase**

- In the verify phase, prototype hardware are built. Quantify all of previous work based on test results. By this stage prototype should be ready for testing and more detailed analysis.
- Some tools used in this phase are
  - Highly Accelerated Life Testing (HALT)
  - Accelerated Life Testing (ALT)
  - Test to Failure (Life data analysis) life…etc.
  - Validation usually involves functional and environmental testing on a system to become ready for production.

**Control Phase**

- In control phase it is assured that the process remained unchanged and variation remains with the tolerances.
- Some tools used in control phases are

- Control and charts and process capability studies.
- Human reliability
- Continuous compliance
- Audits, ...etc.

**Design for Environment (DFE)**

- Design for the environment is a design approach to reduce the overall human heat and environment impact of product, processes or services, where impacts are considered through its lifecycle.
- These include basic principles:
    - **Protect the biosphere**: Companies will minimize the release of pollutants that endangers the earth.
    - **Sustainable use of resources**: Companies will use raw materials at a level where they can be sustained.
    - **Reduction and disposal of waste**: Companies will minimize, the waste wherever possible, where waste cannot be avoided, recycling will be adopted.
    - **Wise use of energy**: Companies will use environmentally safe energy and invest in energy conversation.
    - **Risk reduction**: Companies will minimize health risk to employees and the community.
    - **Marketing of safe products/services**: Company of same products and service. Company will market the products that minimize environmental impact and are safe for consumer to use.
    - **Damage compensation**: Companies will take responsibility through clean up and compensation for environmental harm.
    - **Disclosure**: Companies will disclose to employees and community about incident that cause environmental harm or pose health and safety hazards.
    - **Environmental directors**: At least one member of company board will be qualified to represents environmental issues and a senior executive for environmental affairs should be appointed.
    - **Annual audit**: Companies will conduct annual self evaluation of progress in implementing these principles and make results of independent audits available to public.

## Local, Regional and Global Issues

### Local and Regional Issues

- These include the problem of acid rain where production by products in one region can cause acid rain in another region.
- Air pollution and smog also are regional problems.
- Water pollution either in the ground water, river, bay or ocean often caused by herbicides and pesticides, in addition to sub-urban and urban street water run off.
- Acid rain is a regional pollution problem caused by excessive fossil fuels, air emissions for regional area.
- The fuel consumption products are released into the air and cause rain in the surrounding to have a lower acidic pH level, which can cause regional plant and aquatic life to suffer.
- From a product design point of view developing products that use less energy will help reduce this problem.
- Air pollution is a simple problem caused by excessive fossil fuel emission for regional area.
- Nitrogen compounds, CO and $SO_2$ are common problem fuel consumption products that are released into air.
- Other contaminants can enter the environment through water flow streams and landfills as water pollution.

### Global Issues

There are production problems whose manifestations exist on global scales.

- These include concern over, climate change, ozone depletion and biodiversity loss.
- Climate change is a concern over the inevitable consequence of possible large changes in Earth's climate due to increase in green house gases. This issue is mainly because excess in energy usage which through burning of fossil fuels, increase the $\text{CO}_2$ levels in the atmosphere increased. $\text{CO}_2$ level in atmosphere may cause increase in global surface, ocean and atmospheric temperature, which in turn cause drastic climate changes. From a product design point of view, selection of product that uses less energy will help minimize this problem.
- Another global pollution concern is the depletion of ozone layer.
- The ozone layer is the thin layer in the upper atmosphere that blocks most ultraviolet rays from reaching the Earth's surface.
- Fluorocarbon gases, shown to have come from industry may also react with and reduce the ozone gas.

- From product's, go print of developing a product that do not make use of or release these harmful chemicals, either in the manufacture or disposal will help to minimize this problem.
- Minimizing Biodiversity Loss: This is another global environmental impact objective loss for different plant and animal species due to over expanding society can loss of ecosystems.
- Developing products that use less new raw material will help to solve this problem.

## Basic Lifecycle Assessment

- To design a product for the environment, an assessment of environmental impact must be completed.
- The society for toxicology and chemistry (SETAC) has developed a four process for completing a lifecycle assessment.
- This process allows us to understand the environmental impact of the product from manufacture through use through disposal. This process includes following steps.

1. **Goal Definition**: What is the purpose of environmental analysis?
    - A company may wish to reduce CO2 emissions, reduce energy use, reduce component toxicity, etc. The goal should be defined and revised.

      i) Inventory: After defining the goal, the scope of product system should be defined.

        - How much of the manufacturing process will be considered, all the way back to energy consumed and pollution generated when removing raw materials?
        - For what expected customers use pattern should be and typical energy utilization pattern be analyzed?
        - What disposal routes should be considered?
        - Should recycling be assumed?

          ii) Impact assessment: The basic method, ATT's environmentally responsible product assessment method, does not complete numerical lifecycle assessment.

    - Other similar methods include Motorola's product life cycle matrix or the Environmental Impact Factor Analysis (EIFA).

## Weighted Sum Assessment Method

With the second most advanced method, the weighted sum method pre-associates assign numerical impact weighting against the material and process use.

iii) Integration: The assessment must be interpreted for areas of high leverage impact reduction for comparison against other alternatives.

- A more advanced approach is to complete a full life cycle assessment.
- Assess the impact of material usage and waste generation of each stage in the product life cycle. Rather than the full life cycle assessment, however, one might instead inventory of the parts used in product and weight them by average impact weightings.[1]
- That is one might break the product down into quantities of materials in the product by weight.[2]
- Then one might establish environmental impact of different materials by weight and sum this score.[3]
- One such approach was developed in Europe on the initiative of ministry of Housing, spatial planning and environment in Netherland called the Eco-indicator.
- Because the Eco-indicator was developed in Europe, it is based on average European values for the process that describes material, production, treatment, transportation energy generation.
- Therefore the application of the Eco-indicator to Non-European regions will not be entirely appropriate.
- E.g.: Products factor that increase acid rains are over weighted for many regions in U.S.
- The eco indicator system operates by having analyst first establish the mass of component materials in the product, their means of production and the means of disposal.

## Design Failure Mode Effect Analysis

**FMEA** evaluates each process step and assigns a score on a scale of 1 to 10 for the following variables:

1. **Severity**: Assessment the impact of the **failure mode** (the error in the process) with 1 representing the least safety concern and 10 representing the most dangerous safety concern. In most cases processors with severity scores exceeding 8 may require a fault tree analysis which estimates the probability of the failure mode by breaking it down further sub elements.
2. **Occurrence**: Assess the chance of **failure** happening with 1 representing the highest occurrence. For example, a score of 1 may be assigned to a failure that happens once in every 5 years. While a score of 10 may be assigned to a failure that occurs once per hour, once per minute etc.[4]
3. **Detection**: Assesses the chance o[5]f **failure being detected** with 1 representing the highest chance of detection and 10 representing the lowest chance of detection.
4. **RPN: Risk Priority Number**: $\text{Severity} \times \text{Occurrence} \times \text{Detection}$ by rule of thumb any **RPN** value exceeding $80$ requires a corrective action. The corrective action leads to a lower **RPN** number.

A simple explanation of how to complete a process FMEA as follows

i) Form a cross functional team of process owners and operations support personnel with a team leader

ii) Have the team leader define the scope, goals and timeline of completing the FMEA.

iii) As a group, complete a detailed process map.

iv) Transfer the process map for the steps of the FMEA process.

v) Assign severity, occurrence and detection scores to each process step as a team.

## Product Life Cycle Management and Product Data Management

### Introduction to PLM

- In today's challenging global market, enterprises must be innovative to survive in competition.
- Business innovation must occur in all dimensions viz product, process and organization, to improve competitiveness and business performances.
- To differentiate themselves enterprises must capture, manage their intellectual assets.
- This can be accomplished through proper application of a Product Lifecycle Management (PLM) approach that addresses the needs of the extended enterprises.
- PLM is a strategic business approach that helps enterprises achieve its business goals of reducing costs, improving quality and shortening time to market.
- PLM is "a strategic business approach that applies a consistent set of business solutions in suppose of the collaborative creation, management and use of product definition information across the extended enterprises from conceptual to end of life-integrating people, process, business system and information".
- PLM helps companies to:
    - i) Deliver more innovative products and services.
    - ii) Reduce cost, improve quality, shorten time to market while achieving targeted Return On Investment (ROI).

- iii) Establish more comprehensive, collaborative and improved relationships with their customers, suppliers as business partners.

- Product lifecycle management (PLM) refers to a set of computer based tools that has been developed to assist a company to more effectively perform the product design and manufacturing functions from conceptual design to product retirement.

Three Major Subsystems to PLM

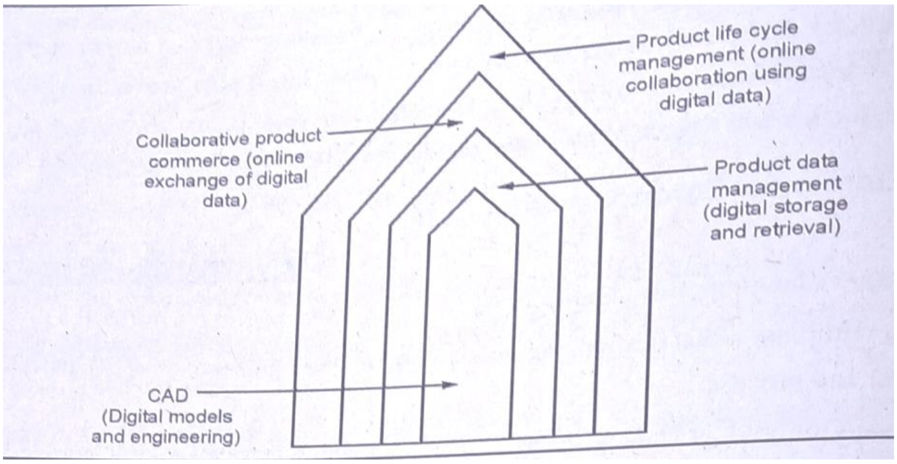
i) Product Data Management (PDM) * a) It provides a link between product design and manufacturing. * b) It provides control of design databases (CAD models, drawings, BOM, etc.) * c) In terms of check in and check out of the data to multiple users, carrying out engineering design changes and control of the release of all versions of component and assembly designs, PLM provides control on all these. * d) Because data security is provided by the PDM system, it is possible to make the design data available electronically to all authorized users along the product development chain. ii) Manufacturing Process Management (MPM): * MPM bridges the gap between product design and production design and production control. It includes such technologies as Computer Aided Process Planning (CAPP), computer aided manufacturing (NC machining and direct numerical control) and computer aided quality assurance (FMEA, SPC, etc.) * It also includes production planning and inventory control using materials requirement planning software. (MRP and MRPII) iii) Customer Relationship Management (CRM): * CRM provides integrated support to marketing, sales and customer service functions. It provides automation of the basic customer contact needs in these functional areas but it also provides analytical capabilities for the data collected from customers to provide information on such issues as market segmentation, measure of customer satisfaction and degree of customer retention.

**Evolution and (Elements) Components of PLM**

- There are four stages of evolution leading to PLM.
- The first phase is computer aided design, engineering and manufacturing (CAD/CAE/CAM) which involves digital models of product and process.
- First phase followed by Product Data Management (PDM) which involves digital storage and retrieval of data.
- PDM in turn led to collaborative product commerce (CPC) which involved on-line exchange of digital data (drawing, marking and on-line catalogues).
- The visual product comprises a digital assembly of its part models. The parts are modelled in 3D using Computer Aided Design (CAD) programs

and saved in standard formats (e.g. IGES and STEP) for exchange between different programs.

- Computer aided engineering (CAE) programs enable simulating the product mechanism and optimizing the shape of each part under static/dynamic loads by simulating the internal stresses.
- The part model can be sent to rapid prototyping (RP) system for automatic fabrication of physical replica for form fit and function testing.
- The tooling models (moulds, dies jigs and fixtures) can be quickly developed by modifying the corresponding parts models.
- Computer aided manufacturing (CAM) enables planning, simulation and optimization of process parameters.
- Finally computer aided inspection systems enable automatic comparison of virtual and real parts for quality assurance.
- The 3D model is the connecting link in various CAX programs (X= design engineering manufacture and inspection). The programs generate huge amount of data which includes the solid model of different iterations and previous versions of products as well as tooling materials, process plans and results of analysis. Other important data includes the bill of material and engineering change order. This necessitates a systematic approach to data storage, verification and retrieval which is achieved by a product data management (PDM) systems.
- The PDM system have rapidly evolved over the last decade and now allow distributed storage and remote access over internet. They provide better data translation for exchange among various programs. Security is handled by data encryption during exchange and facilities for setting limits of access by non core team members. Two are more teams members can synchronise their computer displays, point out product features (by an arrow visible to all team members) and discuss various improvements using a messaging or video conferencing facility.
- Online catalogues of parts made it easy to select, check compatibility with other designed parts and place an order for the part over the internet. This was referred to as collaborative product commerce (CPC).




**Link between Product Data and Product Workflow**

- All the activities along the product workflow create and/or use product data.
- The workflow exist to provide the product data necessary to produce, use and support the product.
- Without product data there would be no need for the product workflow.
- Product data and product workflow are very closely linked.
- Each step or activity in workflow has it's own information needs, information input and information output.
- Within an activity people use information and if information is not available, it may not be possible to complete the activity.
- When change occurs in the workflow corresponding information is produced. Information flow has to be synchronised with workflow so that the information is available when and where it is needed.
- Within individual engineering activity the percentage of time that individuals spend looking for or transferring information is high for many it may be 30 % or even 50 %.
- As time may also be taken up by management tasks, the time actually spent on the functional activity may only be about 30 %.
- In addition, in many companies there are staff who spend 100% of their time looking for information that should have been transmitted by other departments.
- In smallest organisations, product data is created to be used by someone else.
- Once created it should be moved on it should flow to the activity that is going to use it.

- Since product data and product workflow are closely linked, it is not possible to control one without becoming involved with other.

**Product Workflow**

- **Product workflow** is the flow of work through the activities that create or use product data.
- Product workflow starts with initial product specification, includes product use by the customer and ends with the product retirement and recycling.
- The product workflow is rarely linear starting with one well determined activity and continuing serially through other well determined activity until it reaches a well determined final activity.
- The product workflow depends on many parameters such as product/market features, cost time, resources, number and frequency of iterations, number and frequency of changes.
- The product workflow in a particular company will depend greatly on the type of product and the company's position in the value chain.
- Within process industry product workflow for a producer of fine chemicals will differ from that of producer of bulk chemicals.
- Similarly, in discrete manufacturing industries the product workflow will be different for producers of airplanes, machine tools and durable consumer goods.
- The incurred cost of changes during conceptual design is negligible compared to that of changes after the design is released.
- When design changes occurs late in the product workflow they can be extremely expensive, impacting the product production equipment and the launch date.
- Unlike the flow of materials on the shop floor, for which the workflow is defined in details many parts of product workflow are poorly defined and poorly structured.
- The lack of product workflow structure makes it difficult to manage the time and cost of a given project.
- When product workflow is brought under control, lead times are reduced, quality goes up and costs go down.

**Product Data Management**

- Product Data Management (PDM) or Product Information Management (PIM) is the business function within product lifecycle management is responsible for the management and publication of product data.
- The goal of PDM includes ensuring all stakeholders share a common understanding.
- PDM is also the use of software to manage product data and process related information in a single, central system.
- This information includes CAD data, models, part informations, manufacturing instructions, requirements, notes and documents.
- The ideal PDM system is accessible by multiple applications and multiple teams across an organisation and support business specific needs.
- Within PDM the focus is on managing and tracking the creation, change and archive all of the information related to a product.
- The information being stored and managed on one or more file server will include engineering data such as CAD models, drawing and their associated documents.
- PDM serve as central knowledge repository for process and product history and promotes integration and data exchange among all business users who interact with products including projects managers, sales people, buyers and quality assurance team.
- The central database also manage 'metadata' such as author of file and release status of the components. This control check in and check out of the product data to multi user, carry out engineering change management and release control on all issues of components in product, built and manipulate the product structure bill of materials and assembly.
- PDM is focused on capturing and maintaining information on products and/or services through it's development.
- Typical information managed in PDM module include
    - Part name
    - Part description
    - Supplier
    - Vendor part number and description
    - Unit of measure
    - Price

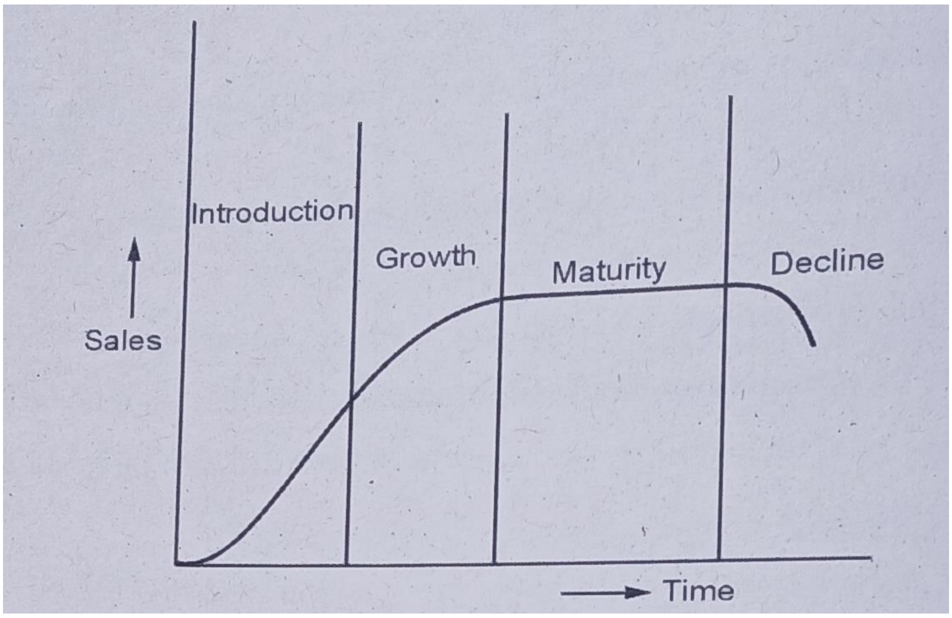
- Schematic or CAD drawings
- Material data sheet

**Benefits of PDM**

- Find the correct data quickly
- Improve productivity and reduce cycle times
- Reduce development errors
- Meet business requirement.

**Product Life Cycle and Phases**

- A product has a life of it's own and goes through cycle. Although different products have different types of life cycles, traditional product life cycle for most product is shown in Fig.
- Product typically goes into four phases during lifetime and each phase is different and requires marketing strategies unique to the stage.




**Introduction stage**

- This stage involves introducing a new and previously unknown product to buyers. Sales are small production process is new and cost reduction through economics of size or experience curve have not been realized. The promotion plan is geared to acquainting buyers with the product.

**Growth stage**

- In this stage sales go rapidly. Buyers have become acquainted with the product and are willing to buy it. So new buyers enters the market and previous buyers come back as repeat buyers. Production may need to be ramped up quickly and may require a large infusion of capital and expertise into the business. Cost reduction occurs as the business moves down the experience curve and economics of size are realized. Profit margins are often large.

- Competitors may enter the market but little rivalry exists, because the market is growing rapidly.

**Maturity stage**

- In this stage the market becomes saturated, production has caught up with demand and demand growth slows. There are few first time buyers most buyers are repeat buyers. Competition becomes intense, leading to aggressive promotional and pricing programs to capture market share from competitors or just to maintain market share. Although experience curve and size economies are achieved. Intense pricing programs often lead to smaller profit margins. Although companies try to differentiate their products the products actually becomes more standardized.

**Decline stage**

- In this stage buyers move on to other products and sales drop. Intense rivalry exists among competitors. Profits dry up because of narrow profit margins and declining sales some businesses leave the industry. The remaining businesses try to receive interest in the product. If they are successful, sales may begin to grow. If not, sales will stabilize or continue to decline.

**Life Cycle of Bajaj Scooters**

**Introduction Phase**

- The Bajaj group was founded in 1926 by Jamnalal Bajaj. In the mid-1940s, Bajaj Auto Limited (BAL) started as an importer of two- and three-wheelers.

- In 1959, the company secured a license from the Government of India (GoI) to manufacture two- and three-wheelers.

- In 1960, BTCL was renamed Bajaj Auto Ltd. and in the same year it entered into a technical collaboration with Piaggio for the manufacture of scooters.

- With its collaboration with Piaggio coming to an end in the early 1970s, BAL started manufacturing scooters under the Bajaj brand.

**Growth Period**

- There was a time when people had to wait a lot of a time after pre-booking their Chetak Scooter.

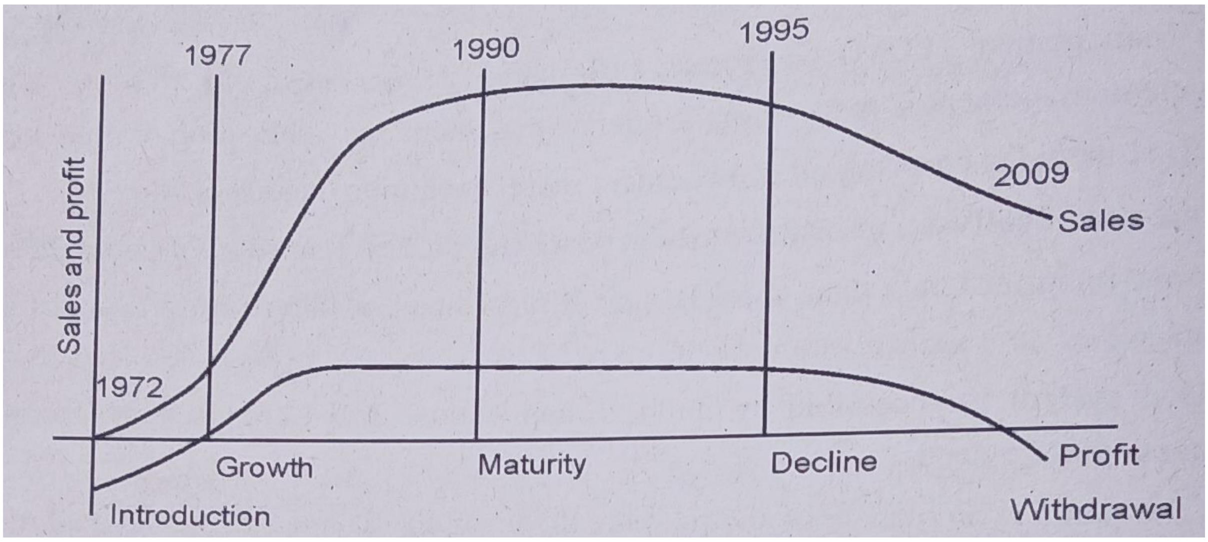
- The launch of Bajaj Chetak Scooter was mainly targeted at economical class and people could afford it considering its packed competition in the same segment market.
- Late 70s had been the golden period for the Bajaj scooters, especially the Chetak scooter.
- In 1977 Bajaj Auto claimed to have sold a lakh of Chetak scooters in just one financial year.
- The main marketing strategy used by Bajaj Chetak scooter was mainly targetting the emerging middle class in India and adding the feeling of 'we' as the feeling of belongingness by strong sentimental slogans like "Hamara Bajaj".
- The approach of this marketing strategy adopted by Chetak scooter was non-pragmatic.
- Over 2 and a half decades until the half of 1995 the company did not face any stiff competition from any 2 wheeler market segment, apart from the introduction of minor changes in its performance parameters meaning more augmented changes in the scooter feel-like colour, some peripheral outer changes in parts.
- Some of the competition was from the scooter segments LML Vespa Scooters.

**Maturity Period**

- In the late 1990s, the Indian two-wheeler market witnessed a shift in consumer preferences. The popularity of geared scooters began to wane while that of motorcycles soared.
- A tough body, low maintenance and initial cost and good resale value are the key attributes that are related the Chetak scooter bearing trust seal from the house of Bajaj.
- There were various reasons for the shift: India was undergoing a demographic change, with the proportion of younger people in the population growing significantly; the economy was growing, which increased the disposable incomes of the middle class; also, many newer models of motorcycles, with improved designs and modern technology had become available in the market.
- While these changes were taking place in the market, the features of scooters, especially those of the Bajaj Chetak scooter, remained essentially unchanged.

**Decline Period**

- The primary reason is that the Brand forgot the customers. The company failed to understand the changing perception of the customers towards scooters.
- Rather than looking at the customers, the company focused on influencing Government to block the opening up of economy. Bajaj never did anything with the product.
- For 40 years, Chetak had the same look, same quality and style.
- During the mid-nineties the company realised lately that the segment has shifted to motorcycles. Scooters were no longer the option.




**Review Questions**

1. Discuss product lifecycle management and data management.
2. Discuss various elements of product lifecycle in detail.
3. Elaborate the concept of PLM.
4. Discuss the concept of product data and product workflow in PLM.
5. Explain the product life cycle phases and corresponding technology.
6. Discuss customer involvement and vendor development in PCM.
7. Discuss the role of people tools, process, methods and data in PLM.
8. Discuss product data management.